# Plant-Caterpillar-Bee Population Models

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## 1 Stage 1 Caterpillar-Plant Model

### 1.1 Assumptions

1. In the absence of caterpillars, plants grow logistically with intrinsic growth rate $r$ and carrying capacity $K$.
2. Plant loss due to herbivory (caterpillar feeding) is proportional to the product of plant and caterpillar abundances, with coefficient $a$.
3. Plant losses are converted into caterpillar population growth with efficiency $\varepsilon$.
4. Caterpillars have a constant natural mortality rate $\mu$.

### 1.2 Variables and Units

| Variable | Meaning | Units |
|---|---|---|
| $P$ | Plant abundance | Plants |
| $C$ | Caterpillar abundance | Caterpillars |
| $t$ | Time | Days |

### 1.3 Parameters and Units

| Parameter | Biological meaning | Units |
|---|---|---|
| $r$ | Intrinsic plant growth rate | $\text{day}^{-1}$ |
| $K$ | Plant carrying capacity | Plants |
| $a$ | Rate of plant loss due to caterpillar feeding | $(\text{caterpillar} \cdot \text{day})^{-1}$ |
| $\varepsilon$ | Efficiency of converting plant losses into caterpillar population growth | caterpillars/plant |
| $\mu$ | Natural caterpillar mortality rate | $\text{day}^{-1}$ |

### 1.4 Model Equations

$$\begin{aligned} \frac{dP}{dt} &= rP\left(1 - \frac{P}{K}\right) - aPC, \\ \frac{dC}{dt} &= \varepsilon aPC - \mu C. \end{aligned} \tag{1}$$

# 2 Stage 2 Caterpillar Plant Bee Model

## 2.1 Assumptions

1. Plant growth, caterpillar feeding, and caterpillar mortality follow the Stage 1 model.
2. Flower visitation per bee increases with plant abundance and saturates at $v$, reaching half its maximum when $P = H$.
3. Flower visits support plant growth and bee population growth. Additional plant growth from bees is limited by the plant carrying capacity $1 - P/K$.
4. Bees experience natural mortality and density-dependent competition (as bees increase, competition between them increases as flower's resources deplete).

## 2.2 Variables and Units

| Variable | Meaning | Units |
|---|---|---|
| $P$ | Plant abundance | Plants |
| $C$ | Caterpillar abundance | Caterpillars |
| $B$ | Bee abundance | Bees |
| $t$ | Time | Days |

## 2.3 Parameters and Units

| Parameter | Biological meaning | Units |
|---|---|---|
| $r$ | Intrinsic plant growth rate | $\text{day}^{-1}$ |
| $K$ | Plant carrying capacity | plants |
| $a$ | Rate of plant loss due to caterpillar feeding | $(\text{caterpillar} \cdot \text{day})^{-1}$ |
| $\varepsilon$ | Efficiency of converting plant losses into caterpillar population growth | caterpillars/plant |
| $\mu$ | Natural caterpillar mortality rate | $\text{day}^{-1}$ |
| $v$ | Maximum flower visits per bee | $\text{visits}/(\text{bee} \cdot \text{day})$ |
| $H$ | Plant abundance at which visitation per bee reaches half its maximum | plants |
| $\beta$ | Additional plant growth per bee flower visit | plants/visit |
| $\gamma$ | Bee population growth contribution per flower visit | bees/visit |
| $d$ | Natural bee mortality rate | $\text{day}^{-1}$ |
| $q$ | Strength of density-dependent bee competition | $(\text{bee} \cdot \text{day})^{-1}$ |

### 2.4 Model Equations

$$
\begin{aligned}
\frac{dP}{dt} &= rP\left(1-\frac{P}{K}\right) - aPC + \beta\frac{vPB}{H+P}\left(1-\frac{P}{K}\right), \\
\frac{dC}{dt} &= \varepsilon aPC - \mu C, \\
\frac{dB}{dt} &= \gamma\frac{vPB}{H+P} - dB - qB^2.
\end{aligned}
\tag{2}
$$